Notes: Guo (RFS, 2025)
Earnings Extrapolation and Predictable Stock Market Returns

Contents

1 Big picture

- **Question.** Can a simple timing pattern tied to the quarterly earnings cycle make aggregate market returns predictable even though unconditional monthly autocorrelation is close to zero?
- **Main empirical pattern.** U.S. market returns in the first month of a quarter—January, April, July, and October—predict future returns with different signs depending on whether the future month is also a first month of a quarter.
- **Mechanism.** Investors extrapolate announced earnings but compress the true step-like earnings-predictability function into a smoother decay function.
- **Return predictability strength.** Real-time forecasts based on the paper's signal generate monthly out-of-sample R^2 values around 3.6%–4.3%.
- **Economic takeaway.** Market-wide return predictability can arise from how investors process the timing of earnings announcements, not only from traditional time-varying risk premia.

2 Data and measurement

2.1 Newsy and non-newsy months

Newsy months are January, April, July, and October. They are special because the U.S. quarterly earnings cycle makes these months the early months of the main announcement seasons.

Interpretation: The newsy-month classification is mechanical and tied to reporting institutions.

2.2 Aggregate returns and flagship signal

The main return sample uses CRSP monthly market returns from 1926 to 2021. The flagship signal is the sum of returns in the past four newsy months:

$$S_t = \sum_{j=1}^{4} mkt_{nm(t,j)}.$$

Interpretation: Four newsy months correspond roughly to a one-year lookback window, aligning the paper with the momentum literature.

2.3 Survey and industry data

IBES detail data are used to construct firm-level and aggregate earnings surprises. Industry returns are value-weighted and measured in excess of the aggregate market return.

Interpretation: Survey expectations test the belief channel directly, while industry returns test whether information transfer works within tighter economic groups.

3 Models and empirical specifications

3.1 Baseline lead-lag regression

$$mkt_t = \alpha + \sum_{j=1}^{4} \beta_j mkt_{nm(t,j)} + \epsilon_t.$$

Interpretation: The regression asks whether past newsy-month returns forecast current market returns.

3.2 Newsy-month interaction regression

$$mkt_t = \alpha + \beta_1 S_t + \beta_2 S_t I_t^{nm} + \beta_3 I_t^{nm} + \epsilon_t.$$

Interpretation: A negative β_2 means that the same signal predicts weaker or negative returns when the dependent month is newsy.

3.3 Out-of-sample forecasting

$$R^2_{OOS} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t (r_t - \bar{r}_t)^2}.$$

Interpretation: Positive R^2 means the forecast beats the expanding-window historical mean.

3.4 Parameter-compression model

Investors believe future earnings relevance decays smoothly. In reality, predictive power drops discretely in newsy months. The forecast error is the difference between the compressed smooth belief and the true step-like loading.

$$db_{t+1} = x_t^{exp} + u_{t+1}.$$

Interpretation: A single extrapolative rule generates both underreaction and overreaction because the relevance of the same signal differs across forecasted months.

4 Identification and empirical design

The paper's credibility comes from six features: mechanical earnings-calendar timing, fixed lag definitions, real-time forecasting tests, survey expectation evidence, industry-level heterogeneity tests, and international evidence. The main limitation is that the paper is not a randomized experiment; return predictability could also be affected by trading frictions, seasonality, or other behavioral forces.

5 Tables and figures

5.1 Table 1: The U.S. earnings cycle—when do firms end fiscal quarters and announce earnings?

Read: 85.6% of fiscal quarters end in March, June, September, or December. For aligned and timely announcements, 46.5% are in January/April/July/October and 46.3% are in February/May/August/November.

Economic message: The table establishes the institutional source of the newsy-month classification.

Table 1
The U.S. earnings cycle: When do firms end their fiscal quarters, and when do they announce earnings?

	A: FQ end month		*B: Earnings announcement month*			
			All announcements		Aligned & timely	
	Count	Percent	Count	Percent	Count	Percent
Group 1: Jan/Apr/Jul/Oct	78,747	8.9%	387,466	43.6%	345,986	46.5%
Group 2: Feb/May/Aug/Nov	49,385	5.6%	385,806	43.4%	344,666	46.3%
Group 3: Mar/Jun/Sep/Dec	761,310	85.6%	116,170	13.1%	53,830	7.2%
Total	889,442	100.0%	889,442	100.0%	744,482	100.0%

Panel A counts the company-fiscal quarters in Compustat by the months in which they end. The fiscal quarters always end on the last day of a month, and 85.6% means that 85.6% of the fiscal quarters end on the last day of March, June, September, or December. Panel B counts the company-fiscal quarters by the months in which the fiscal quarters' earnings are announced. The "All announcements" columns include all data present in panel A. The "Aligned & timely" columns apply two filters: i) the fiscal quarter ends in the Group 3 months and ii) the earnings announcement occurs within 92 days of the fiscal quarter end date. Data are quarterly from 1971 to 2021.

Table 1: The U.S. earnings cycle.

5.2 Table 2: Lead-lag relations of U.S. monthly market returns and Figure 1: Timing of variables

Read: Table 2 shows that past newsy returns predict reversal in newsy months and continuation in non-newsy months. Figure 1 clarifies the moving definition of $mkt_{nm(t,j)}$.

Economic message: This is the core empirical fact: the same signal changes sign with the timing of the dependent month.

Table 2
Lead-lag relations of U.S. monthly market returns

	(1) All	(2) NM	(3) Non-NM	(4) Difference
$mkt_{nm(t,1)}$	0.090*	–0.168**	0.219***	–0.388***
	[1.73]	[–2.32]	[3.92]	[–4.23]
$mkt_{nm(t,2)}$	0.014	–0.177***	0.109**	–0.287***
	[0.34]	[–2.85]	[2.29]	[–3.66]
$mkt_{nm(t,3)}$	0.069	0.041	0.083**	–0.042
	[1.55]	[0.44]	[2.07]	[–0.41]
$mkt_{nm(t,4)}$	0.027	–0.086	0.084**	–0.169**
	[0.69]	[–1.26]	[2.09]	[–2.14]
const	0.007***	0.017***	0.002	0.015***
	[3.40]	[4.77]	[0.86]	[3.53]
N	1,136	378	758	1,136
R-sq	0.013	0.064	0.072	0.070

Column 1 of this table reports results from the following monthly time-series regression: $mkt_t = \alpha + \sum_{j=1}^{4} \beta_j mkt_{nm}(t,j) + \epsilon_t$. Here, mkt_t is the U.S. aggregate market return in month t, and $mkt_{nm}(t,j)$ is the return in the jth newsy month (Jan, Apr, Jul, Oct) preceding month t. Column 2 reports the same regression on the subsample in which the dependent variable is returns of the newsy months. Column 3 is for the non-newsy months. Column 4 reports the difference between the coefficients in columns 2 and 3, extracted from this regression, $mkt_t = \alpha + \sum_{j=1}^{4} \beta_j mkt_{nm}(t,j) + \sum_{j=1}^{4} \gamma_j mkt_{nm}(t,j) \times I_t^{nm} + \delta I_t^{nm} + \epsilon_t$, where I_t^{nm} is a dummy variable taking the value of 1 when month t is newsy. Data are monthly from 1926 to 2021. T-statistics computed with White standard errors are reported in square brackets. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

months, giving rise to the "street" notion of earnings seasons. It is worth noting that the newsy months are special for the purpose of inferring aggregate cash flows. If one is interested in a firm's idiosyncratic cash flow, the corresponding earnings announcement is equally informative whether or not it occurs in the newsy months. Therefore, the first-order implications the newsy months have are for the aggregate market.

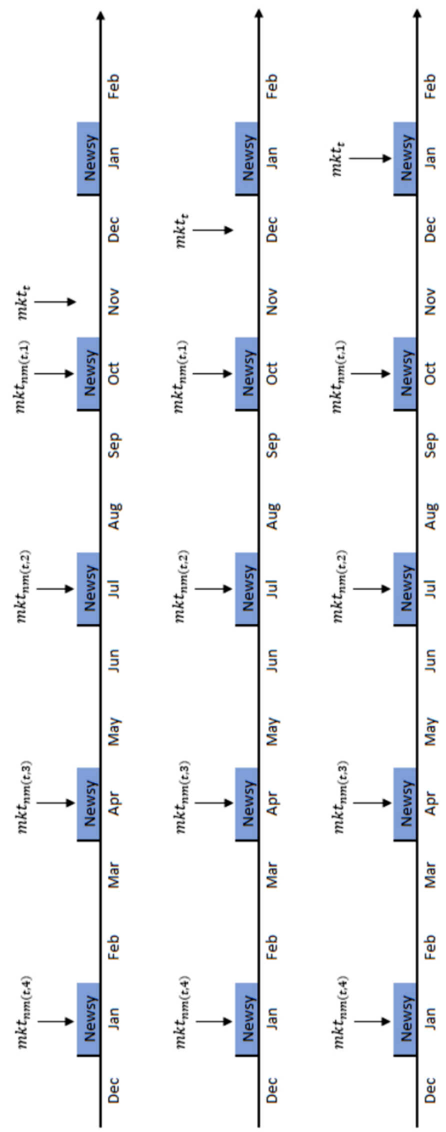

Figure 1
Timing of independent and dependent variables in the U.S. market return forecasting regression
This figure shows how the independent variables, $mkt_{nm(t,j)}$, in the U.S. return forecasting regression $mkt_t = \alpha + \sum_{j=1}^{4} \beta_j mkt_{nm(t,j)} + \epsilon_t$ progress as the dependent variable moves forward. Notice $mkt_{nm(t,j)}$ is the jth most recent newsy month strictly before month t.

(a) Table 2: U.S. lead-lag return regressions.

(b) Figure 1: Timing of variables.

5.3 Table 3: Lead-lag relations across subsamples

Read: The four-newsy-month signal predicts newsy months negatively and non-newsy months positively; the interaction coefficient is -0.232 and highly significant.

Economic message: The conditional sign flip is stable across historical subsamples.

Table 3
Lead-lag relations of U.S. monthly market returns, across subsamples

	(1) All	(2) NM	(3) Non-NM	(4) All	(5) Post-WWII	(6) First Half	(7) Second Half
$\sum_{j=1}^{4} mkt_{nm(t,j)}$	0.052**	–0.103***	0.129***	0.129***	0.082***	0.162***	0.090***
	[2.04]	[–2.67]	[4.76]	[4.76]	[3.59]	[3.86]	[3.17]
$\sum_{j=1}^{4} mkt_{nm(t,j)} \times I_t^{nm}$				–0.232***	–0.157***	–0.279***	–0.176***
				[–4.93]	[–3.51]	[–3.96]	[–3.09]
I_t^{nm}				0.016***	0.011***	0.019***	0.012**
				[3.51]	[2.68]	[2.68]	[2.26]
const	0.007***	0.017***	0.002	0.002	0.005**	–0.001	0.005
	[3.30]	[4.60]	[0.74]	[0.74]	[2.16]	[–0.19]	[1.56]
N	1,136	378	758	1,136	893	567	569
R-sq	0.008	0.029	0.057	0.047	0.026	0.061	0.032

Column 1 of this table reports results from the following monthly time-series regression: $mkt_t = \alpha + \beta \sum_{j=1}^{4} mkt_{nm(t,j)} + \epsilon_t$. Here mkt_t is the U.S. aggregate market return in month t, and $mkt_{nm(t,j)}$ is the return in the jth newsy month (Jan, Apr, Jul, Oct) preceding month t. Column 2 reports the same regression as in 1, but on the subsample in which the dependent variable is returns of the newsy months. Column 3 is for the non-newsy months. Column 4 reports the results of $mkt_t = \alpha + \beta_1 \sum_{j=1}^{4} mkt_{nm(t,j)} + \beta_2 \sum_{j=1}^{4} mkt_{nm(t,j)} \times I_t^{nm} + \beta_3 I_t^{nm} + \epsilon_t$, where I_t^{nm} is a dummy variable taking the value of 1 when month t is newsy. Columns 5–7 report results for the regression in column 4 on the subsamples of the post-WWII period (1947–2021), the first half (1926–1973), and the second half (1974–2021). T-statistics computed with White standard errors are reported in square brackets. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

Table 3: Subsample evidence.

5.4 Table 4: Real-time forecasts and Figure 2: Expected returns and Treasury rates

Read: Table 4 reports out-of-sample R^2 values around 3.6%–4.3% for the paper's signal. Figure 2 shows high-frequency expected-return variation that sometimes falls below the Treasury bill rate.

Economic message: The signal is economically meaningful and challenges the assumption that expected returns are persistent and always positive relative to bills.

Table 4
Predicting U.S. aggregate market returns with real-time coefficients

Method	0	1	2	3	4	5	6	7
R^2	0.49%	3.72%	3.60%	3.68%	3.97%	3.77%	3.82%	4.30%

The R^2 in this table are calculated as $1 - \frac{\sum_{t=1}^{n}(r_t - \hat{r_t})^2}{\sum_{t=1}^{n}(r_t - \overline{r_t})^2}$, where $\overline{r_t}$ is the expanding window mean of past stock returns and $\hat{r_t}$ is the forecast being evaluated. This R^2 is positive only when the forecast outperforms the expanding window mean of past stock returns. Method 0 comes solely from Campbell and Thompson (2008) and functions as a benchmark. It is the valuation constraint + growth specification with fixed coefficients. A simple average is taken from the forecasts based on dividend/price, earnings/price, and book-to-market ratios, motivated by Huang et al. (2020). Method 1 uses the signal that is simply the sum of past 4 newsy month returns. The coefficients are extracted from simple expanding-window OLS of past returns on past signals, separately for newsy and non-newsy month dependent variables. The signal used in methods 2–7 is the sum of the past 4 newsy month returns, subtracting its expanding window mean, and sign flipped if the dependent variable is a newsy month. Method 2 uses the same coefficient estimation method as in method 1. Method 3 replaces the constant terms with the expanding window means of past newsy and non-newsy month returns. Method 4 replaces the constant terms with the forecast in method 0. Methods 5–7 are methods 2–4 with the coefficients estimated on the combined sample of newsy and non-newsy months. Data are monthly from 1926 to 2021.

(a) Table 4: Return-forecast R^2.

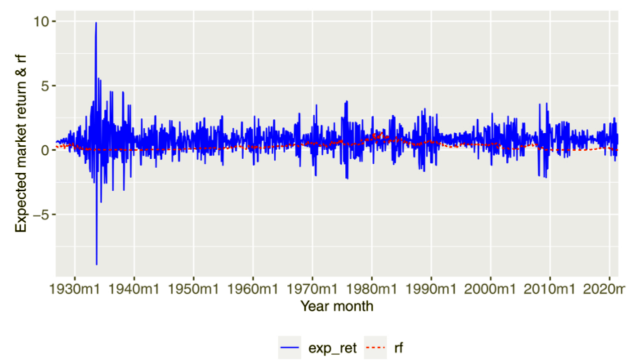

Figure 2
The expected stock market returns and the 1-month Treasury rates
The blue line (exp_ret) of this figure plots the expected stock market return as in method 6 of Table 4. The dashed red line (rf) is the 1-month Treasury rates. The unit is percent per month.

(b) Figure 2: Expected returns and T-bill rate.

5.5 Figure 3: Continuation versus reversal strength across decades

Read: Decades with stronger continuation in non-newsy months tend to have stronger reversal in newsy months.

Economic message: The two arms of predictability appear connected rather than independent anomalies.

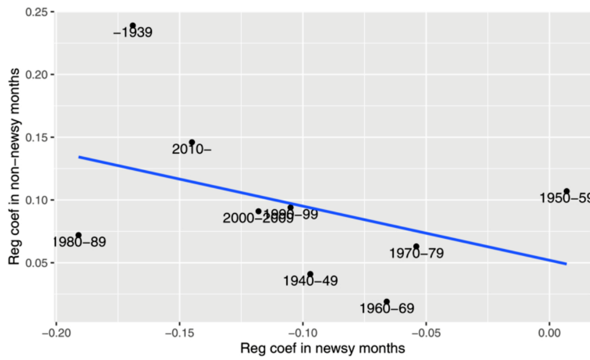

Figure 3
Continuation vs. reversal strength across decades
This figure plots the strength of the continuation arm of the return, represented by β_1 in Table A1 against the strength of the reversal, represented by $\beta_1+\beta_2$. Each decade in the CRSP history is represented by a point. The blue line is fitted across the points.

Figure 3: Continuation versus reversal across decades.

5.6 Figure 4: Predictive power of 0E earnings and Figure 5: E/L earnings loadings

Read: Figure 4 shows true step-like predictive power and smooth investor beliefs. Figure 5 restates the same mechanism as loadings on past earnings shocks.

Economic message: These figures show how parameter compression creates both underreaction and overreaction.

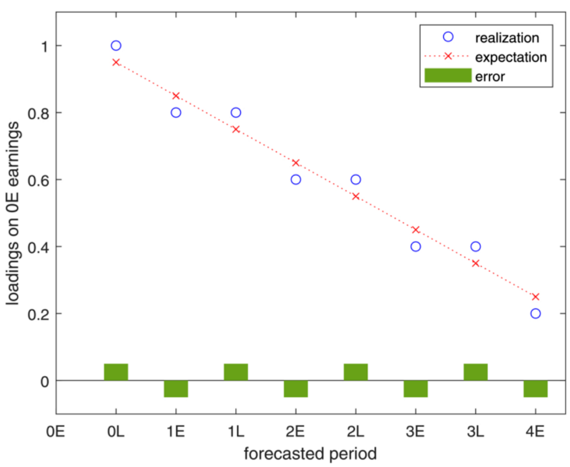

Figure 4
Predictive power of earnings in period 0E on earnings in future periods
This figure is an example of the predictive power of the earnings announced in period 0E on earnings in future periods, namely the regression loadings of these future earnings on 0E earnings. The "E" periods represent the early half of the earnings cycle and are newsy. The "L" periods represent the late half of the earnings cycle and are non-newsy. Time progresses from left to right for eight periods to period 4E. The blue circles represent the actual predictive power across the forecasted periods, following a step function from left to right. The red circles represent the predictive power in investors' belief. The green bars show their differences, representing investors' forecasting errors.

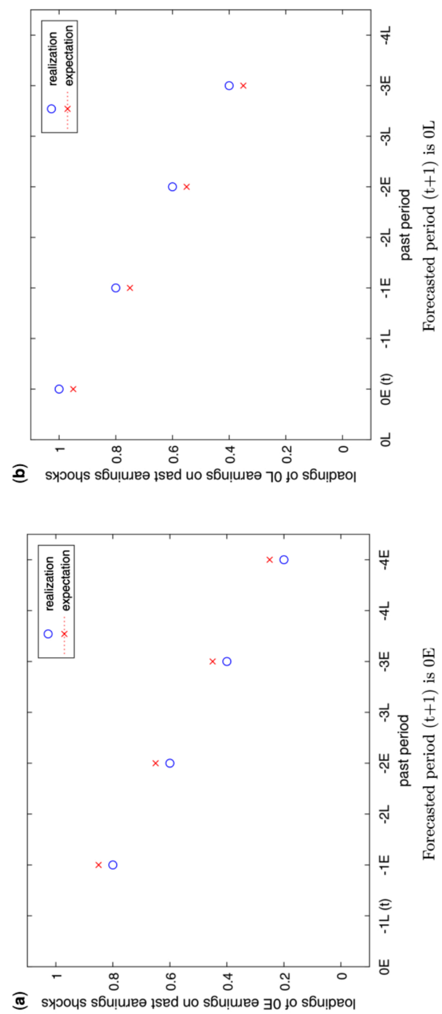

Figure 5
Loadings of E(arly) and L(ate) period earnings on past earnings shocks
The left panel of this figure is an example of the loadings of the earnings announced in period 0E on past earnings shocks. The "E" periods represent the early half of the earnings cycle and are newsy. The "L" periods represent the late half of the earnings cycle and are non-newsy. From left to right, time moves backward for eight periods to period –4E. The blue circles represent the actual earnings' loadings across past periods. The red circles represent expected earnings' loadings across past periods. Complementing the left panel, the right panel of this figure shows the loadings of the earnings announced in period 0L on past earnings shocks.

(a) Figure 4: Step-like predictive power.

(b) Figure 5: E/L loadings.

5.7 Table 5: No reversal in the preseason portion and Table 6: Weak return pattern in Q1

Read: The reversal is absent in the first five trading days of newsy months and is weak in Q1.

Economic message: The timing evidence ties reversal to active earnings-season information rather than generic first-month seasonality.

Table 5
No reversal in the preseason portion of the newsy months

	(1) mkt_t	(2) mkt_t^{FW}	(3) mkt_t^{ExclFW}
$\sum_{j=1}^{4} mkt_{nm(t,j)}$	–0.103***	0.019	–0.120***
	[–2.67]	[1.03]	[–2.92]
const	0.017***	0.005***	0.012***
	[4.60]	[3.35]	[3.38]
N	378	378	378
R-sq	0.029	0.005	0.047

Column 1 of this table reports results from the monthly time-series regression $mkt_t = \alpha + \beta \sum_{j=1}^{4} mkt_{nm(t,j)} + \epsilon_t$ on the subsample in which the dependent variable is the first month of a quarter, or Jan, Apr, Jul, Oct. Here mkt_t is the market return in month t, and $mkt_{nm(t,j)}$ is the market return in the jth newsy month preceding month t. Column 2 replaces the dependent variable with mkt_t^{FW}, which is the aggregate market return in the first 5 trading days in month t. Column 3 has the dependent variable mkt_t^{ExclFW}, which is the aggregate market return in month t excluding the first 5 trading days. Data are monthly from 1926 to 2021. T-statistics computed with White standard errors are reported in square brackets. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

Table 6
Weak return pattern in Q1

	Q2, Q3, and Q4			Q1		
	(1) First months	(2) Second months	(3) Third months	(4) January	(5) February	(6) March
$\sum_{j=1}^{4} mkt_{nm(t,j)}$	–0.125***	0.173***	0.121***	–0.024	0.056	0.084
	[–2.83]	[3.67]	[2.95]	[–0.35]	[1.07]	[1.14]
const	0.017***	0.003	0.000	0.016***	0.003	0.002
	[3.79]	[0.69]	[0.10]	[2.88]	[0.58]	[0.33]
N	283	284	284	95	95	95
R-sq	0.040	0.088	0.054	0.002	0.015	0.023

This table reports results from the following monthly time-series regression: $mkt_t = \alpha + \beta \sum_{j=1}^{4} mkt_{nm(t,j)} + \epsilon_t$. Here, mkt_t is the U.S. aggregate market return in month t, and $mkt_{nm(t,j)}$ is the return in the jth newsy month (Jan, Apr, Jul, Oct) preceding month t. In columns 1–3, the dependent variables are the first, second, and third months of a quarter, where the quarter is not Q1. In columns 4–6, the dependent variables are returns of January, February, and March. Data are monthly from 1926 to 2021. T-statistics computed with White standard errors are reported in square brackets. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

6.2. Evidence from survey data

6.2.1 Aggregate level evidence In this subsection, I test my theory with

Tables 5 and 6: Preseason and Q1 tests.

5.8 Table 7: Survey evidence from IBES and Table 8: Firm-level evidence from IBES

Read: Aggregate and firm-level IBES evidence both show negative newsy-month interactions.

Economic message: The belief data support the earnings-extrapolation mechanism directly.

Table 7
Survey evidence from IBES

	(1) All	(2) NM	(3) Non-NM	(4) All	(5) All
$\sum_{j=1}^{12} Sup_{t-j}$	0.061***	0.049***	0.068***	0.068***	
	[8.54]	[4.82]	[9.96]	[9.95]	
$\sum_{j=1}^{12} Sup_{t-j} \times I_t^{nm}$				–0.019**	
				[–2.18]	
$\sum_{j=1}^{4} Sup_{nm(t,j)}$					0.227***
					[10.81]
$\sum_{j=1}^{4} Sup_{nm(t,j)} \times I_t^{nm}$					–0.064**
					[–2.26]
I_t^{nm}				0.001***	0.001***
				[7.71]	[6.12]
const	–0.000	0.000***	–0.000***	–0.000***	–0.001***
	[–1.08]	[3.62]	[–3.90]	[–3.87]	[–5.91]
N	419	140	279	419	419

Column 1 of this table reports results from the following monthly time-series regression: $Sup_t = \alpha + \beta \sum_{j=1}^{12} Sup_{t-j} + \epsilon_t$. Here, Sup_t is the aggregate earnings surprise in month t, which is the market capitalization weighted average of firm-level earnings surprises of all firms announcing in month t. The firm-level earnings surprises are computed as the difference between realized EPS and expected EPS divided by the firm's stock price at the end of the previous month. Column 2 reports the same regression as in 1, but on the subsample in which the dependent variable is aggregate surprises of the newsy months. Column 3 is for the non-newsy months. Column 4 reports the difference between the coefficients in columns 2 and 3, extracted from a regression with additional interaction terms to I_t^{nm}, which is a dummy variable taking the value of 1 if month t is newsy. Column 5 performs the same regression as in column 4, except it additionally confines the surprises on the right-hand side to be those in newsy months. Data are monthly from 1985 to 2021. T-statistics computed with Newey-West standard errors are reported in square brackets. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

(a) Table 7: Aggregate IBES evidence.

Table 8
Firm-level evidence from IBES

	(1) $Sup_{i,t}$	(2) $Sup_{i,t}$	(3) $Sup_{i,t}$
$\sum_{j=1}^{12} Rev_{i,t-j}^{t}$	0.339***		
	[2.87]		
$\sum_{j=1}^{12} Rev_{i,t-j}^{t} \times I_t^{nm}$	–0.222***		
	[–2.70]		
$\sum_{j=1}^{4} Rev_{i,nm(t,j)}^{t}$		0.619***	
		[3.02]	
$\sum_{j=1}^{4} Rev_{i,nm(t,j)}^{t} \times I_t^{nm}$		–0.486***	
		[–3.06]	
$\sum_{j=1}^{8} Rev_{i,nnm(t,j)}^{t}$			0.366***
			[2.98]
$\sum_{j=1}^{8} Rev_{i,nnm(t,j)}^{t} \times I_t^{nm}$			–0.127
			[–1.48]
I_t^{nm}	0.000	0.001***	0.000
	[0.44]	[3.73]	[0.57]
const	0.000*	–0.000*	0.000
	[1.82]	[–1.69]	[1.19]
N	269,384	296,496	284,886

Column 1 of this table reports results from the following stock-month level panel regression: $Sup_{i,t} = \alpha + \beta_1 \sum_{j=1}^{12} Rev_{i,t-j}^{t} + \beta_2 \sum_{j=1}^{12} Rev_{i,t-j}^{t} \times I_t^{nm} + \beta_3 I_t^{nm} + \epsilon_{i,t}$. Here, $Sup_{i,t}$ is the earnings surprise of firm i in month t, computed as the difference between realized EPS and expected EPS divided by the firm's stock price at the end of the previous month. $Rev_{i,t-j}^{t}$ is firm i's earnings revision in month $t-j$ targeting month t's announcement, computed as the difference in firm i's stock-price-scaled expected EPS between month $t-j$ and $t-j-1$, where the expectations target earnings announced in month t. I_t^{nm} is a dummy variable taking the value of 1 if month t is newsy. Column 2 conducts a similar regression, except that the term $\sum_{j=1}^{4} Rev_{i,nm(t,j)}^{t}$ captures the revisions that occurred in the 4 newsy months preceding month t. In column 3, the term $\sum_{j=1}^{8} Rev_{i,nnm(t,j)}^{t}$ captures the revisions that occurred in the eight non-newsy months preceding month t. Regressions are weighted by firm-level market cap normalized by the total market cap of the quarter. Data are monthly from 1983 to 2021. T-statistics computed with standard errors clustered by quarter are reported in square brackets. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

$Sup_{i,t}$ is the earnings surprise of firm i in month t, computed as the difference

(b) Table 8: Firm-level IBES evidence.

5.9 Table 9: Industry excess returns and Table 10: Heterogeneous effect across industries

Read: The industry return pattern mirrors the aggregate pattern and is stronger in high-connectivity, high-entry-cost, balanced, and large industries.

Economic message: The cross-sectional evidence is a sharp test of the information-transfer mechanism.

Table 9
Lead-lag relations of monthly excess industry returns in the United States, by different industry measures

	(1) All	(2) NM	(3) Non-NM	(4) Difference
A: SIC major group				
$\sum_{j=1}^{4} exret_{i,nm(t,j)}$	0.021***	–0.012	0.038***	–0.050***
	[2.89]	[–0.96]	[4.35]	[–3.26]
N	28,028	9,327	18,701	28,028
R-sq	0.002	0.001	0.007	0.004
B: SIC industry group				
$\sum_{j=1}^{4} exret_{i,nm(t,j)}$	0.020***	–0.008	0.034***	–0.041***
	[2.76]	[–0.64]	[3.81]	[–2.78]
N	24,082	8,037	16,045	24,082
R-sq	0.002	0.000	0.005	0.004
C: SIC industry				
$\sum_{j=1}^{4} exret_{i,nm(t,j)}$	0.020**	–0.017	0.038***	–0.055***
	[2.51]	[–1.39]	[3.84]	[–3.50]
N	19,108	6,353	12,755	19,108
R-sq	0.002	0.001	0.007	0.005

Column 1 of this table reports results from the following industry-month level regression: $exret_{i,t} = \alpha + \beta \sum_{j=1}^{4} exret_{i,nm(t,j)} + \epsilon_{i,t}$. Here, $exret_{i,t}$ is the value-weighted average return of industry i in excess of the market in month t, and $exret_{i,nm(t,j)}$ is the excess return in the jth newsy month (Jan, Apr, Jul, Oct) preceding month t. Column 2 reports the same regression on the subsample in which the dependent variable is returns of the newsy months. Column 3 is for the non-newsy months. Column 4 reports the difference between the coefficients in columns 2 and 3, extracted from this regression: $exret_{i,t} = \alpha + \beta \sum_{j=1}^{4} exret_{i,nm(t,j)} + \gamma \sum_{j=1}^{4} exret_{i,nm(t,j)} \times I_t^{nm} + \delta I_t^{nm} + \epsilon_{i,t}$, where I_t^{nm} is a dummy variable taking the value of 1 when month t is newsy. Panels A to C differ only in the industry variable used. Regressions are weighted by the market cap of industry i as of month $t-1$, normalized by the total market cap of each cross section. Twenty stocks are required in each month-industry. Data are monthly from 1926 to 2021. T-statistics computed with clustered standard errors by month are reported in square brackets. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

(a) Table 9: Industry lead-lag relations.

Earnings Extrapolation and Predictable Stock Market Returns

Table 10
Heterogeneous effect across industries

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Connectivity		Entry cost		Balance		Size		Placebo	
	low	high	low	high	low	high	small	large	missing	random
$\sum_{j=1}^{4} exret_{i,nm(t,j)}$	0.037**	0.052***	0.043**	0.053***	0.039*	0.057***	0.017***	0.027***	0.060	0.001
	[2.35]	[2.85]	[2.02]	[3.00]	[1.95]	[3.40]	[3.30]	[4.57]	[0.76]	[0.18]
$\sum_{j=1}^{4} exret_{i,nm(t,j)} \times I_t^{nm}$	–0.035	–0.121***	–0.036	–0.103***	–0.037	–0.111***	–0.015	–0.028***	0.005	–0.001
	[–1.28]	[–4.03]	[–1.05]	[–3.31]	[–1.00]	[–3.81]	[–1.61]	[–2.84]	[0.05]	[–0.11]
N	7,571	7,290	5,376	4,899	4,830	5,390	250,034	183,316	286	31,613
R-sq	0.004	0.013	0.005	0.011	0.004	0.013	0.002	0.002	0.017	0.000

This table reports results on the following industry-month level panel regression: $exret_{i,t} = \alpha + \beta_1 \sum_{j=1}^{4} exret_{i,nm(t,j)} + \beta_2 \sum_{j=1}^{4} exret_{i,nm(t,j)} \times I_t^{nm} + \beta_3 I_t^{nm} + \epsilon_{i,t}$. Here, $exret_{i,t}$ is the value-weighted return of industry i in excess of the market in month t, $exret_{i,nm(t,j)}$ is the excess return in the jth newsy month (Jan, Apr, Jul, Oct) preceding month t, and I_t^{nm} is a dummy variable indicating whether month t is newsy. Columns 1 and 2 perform the regression on the subsamples in which the within-industry ROE connectivity is respectively low and high. Columns 3 and 4 perform a similar exercise on low and high entry cost industries. Columns 5 and 6 are on unbalanced and balanced industries. Columns 7 and 8 are on industries with a small and larger number of stocks. Column 9 is on an "industry" in which companies have an SIC code of 9999, 9990, 9910, 0, or missing value. Column 10 is on "industries" formed with randomly generated "SIC codes." Twenty stocks are required in each industry-month, except for columns 7–8, where no size filter is applied. Data are monthly from 1971 to 2021 for columns 1–6 and from 1926 to 2021 for columns 7–10. T-statistics computed with standard errors clustered at the month level are reported in square brackets. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$.

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(b) Table 10: Heterogeneity across industries.

5.10 Selected appendix and robustness tables: A2, A3, A4, B1, and C3

Read: The appendix tables benchmark forecasting performance, convert the signal into implementable strategy returns, examine ROE forecasting, and extend the preseason logic internationally.

Economic message: These tests support implementation, accounting-fundamental, and international validity.

Table A2
R^2 in forecasting monthly stock market returns using other signals

	IS	CTOOS10	CTOOS5	OOS10	OOS5	Begin	End
dy	0.68%	−0.28%	−0.28%	−0.48%	−0.48%	1871m2	2019m12
ep	0.88%	0.27%	0.27%	0.28%	0.28%	1871m1	2019m12
bm	1.04%	0.16%	0.07%	−3.02%	−2.93%	1921m3	2019m12
svar	0.25%	0.03%	0.03%	−1.49%	−1.49%	1885m2	2019m12
csp	0.90%	−0.31%	0.24%	−5.81%	−6.12%	1937m5	2002m12
ntis	0.75%	−0.18%	0.02%	−0.13%	−0.78%	1926m12	2019m12
eqis	0.29%	−0.43%	−1.62%	−0.40%	−2.11%	1926m10	2008m4
tbl	0.23%	−0.80%	−0.61%	−1.70%	−1.61%	1920m1	2019m12
lty	0.22%	−1.33%	−1.08%	−1.64%	−1.39%	1919m1	2019m12
dfy	0.49%	−0.22%	−0.22%	−2.45%	−2.24%	1919m1	2019m12
dfr	0.37%	−0.30%	−1.72%	−0.21%	−1.69%	1926m1	2019m12
infl	0.35%	−0.05%	−0.05%	−0.03%	−0.03%	1913m2	2019m12
ik	1.09%	0.30%	0.30%	−0.03%	−0.01%	1947m3	2019m12
cay	0.37%	−0.32%	−0.31%	−0.39%	−0.97%	1947m4	2019m12
tms	0.29%	−0.10%	−0.20%	−1.04%	−1.32%	1920m1	2019m12
aem	0.31%	−2.12%	−2.01%	−40.25%	−38.18%	1947m4	2009m4
hkm	0.60%	−0.09%	0.00%	−0.28%	−0.30%	1970m1	2012m12
rrz	1.45%	−10.18%	−12.30%	−9.83%	−12.00%	1973m1	2014m12
kp	1.05%	0.31%	0.23%	0.36%	0.27%	1930m1	2019m11
svix	0.21%	−5.04%	−6.21%	−7.26%	−8.67%	1996m1	2012m1
gm	0.78%	0.60%	0.36%	−0.52%	−5.20%	1996m1	2012m1
vrp	0.10%	−0.67%	−0.53%	−17.98%	−14.02%	1989m12	2019m12
sent	0.39%	−0.10%	−0.64%	−0.01%	−0.67%	1965m7	2019m12

This table displays R^2 and the sample start/end month of various monthly signals that predict monthly stock market returns in the United States. The "IS" column shows the in-sample R^2s, while the next four columns show the out-of-sample R^2s as in Goyal and Welch (2008). Columns with the 5 and 10 suffixes evaluate the R^2s 5 and 10 years after the date's inception (but not after 1926m1). Columns with the "CT" prefix i) truncate the estimated coefficient on the signal at 0 if it has a different sign from that estimated on the full sample and ii) truncate the forecast at the risk-free rate if it exceeds the risk-free rate, as in Campbell and Thompson (2008). All predictors are sourced from Amit Goyal's website (and follow his naming convention) except for the following: eqis is the new equity issuance share from Baker and Wurgler (2000), aem is the financial factor from Adrian,

(a) Table A2: Forecasting benchmarks.

Table C3
No reversal in the preseason portion of the newsy months: Global evidence

	(1) $mkt_{c,t}$	(2) $mkt_{c,t}^{PS}$	(3) $mkt_{c,t}^{ExclPS}$
$\sum_{j=1}^{4} mkt_{nm(t,j)}$	–0.029	0.011	–0.040**
	[-1.07]	[0.76]	[–2.29]
const	0.017***	0.008***	0.009***
	[3.21]	[2.66]	[2.72]
N	2,305	2,305	2,305
R-sq	0.006	0.002	0.022

Column 1 of this table reports results from the following monthly time-series regression: $mkt_{c,t} = \alpha + \beta \sum_{j=1}^{4} mkt_{c,nm(t,j)} + \epsilon_{c,t}$, on cases in which month t are the first months of the quarters. Here, $mkt_{c,t}$ is the aggregate market return in month t for country c, and $mkt_{c,nm(t,j)}$ is the aggregate market return in the jth newsy "month" (Jan+Feb, Apr+May, Jul+Aug, Oct+Nov) preceding month t for country c. Column 2 replaces the dependent variable with $mkt_{c,t}^{PS}$, the pre-earnings season returns. They are returns in the first 5, 10, or 15 trading days of the month, depending on country c's reporting speed. Column 6 changes the dependent variable to $mkt_{c,t}^{ExclPS}$, which equals $mkt_{c,t} - mkt_{c,t}^{PS}$. Data are monthly from 1964 to 2021. This shorter sample is owing to the availability of the daily return data. T-statistics computed with standard errors clustered at the monthly level are reported in the brackets. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

C.2. Return Predictability Having described the earnings-related information globally, I move on to the return predictability results. My country-level market returns come from Global Financial Data (GFD). Returns are all in U.S. dollars. A number of return series go back further than 1926, but in this analysis I truncate the sample at 1926 to be consistent with the U.S. results.

To start, panel A of Table C2 runs regressions analogous to those in Table 3, except they are now on a panel of country-level aggregate returns, and the newsy "months" now include both the first and second months of each calendar quarter. Column 1 shows an unconditional momentum effect. This reflects country-level momentum in Moskowitz, Ooi, and Pedersen (2012), which is in fact stronger in more recent periods. Columns 2 and 3 then show that this component is much stronger

(b) Table C3: International preseason evidence.

6 Short summary

- Past newsy-month returns predict reversal in future newsy months and continuation in future non-newsy months.
- The two effects offset in unconditional autocorrelations.
- Real-time return forecasts based on the signal have unusually high out-of-sample R^2.
- A behavioral model of earnings extrapolation with parameter compression explains the pattern.
- IBES survey data, firm-level expectations, industry returns, and international evidence support the mechanism.

7 Detailed conclusion

7.1 Core conclusion

Guo shows that aggregate return predictability can be strong but hidden: conditioning on the earnings calendar reveals sign-changing predictability that disappears in unconditional autocorrelations.

7.2 Economic intuition

Investors extrapolate earnings news but compress the true step-like relevance of past earnings into a smoother decay. This creates overreaction into newsy months and underreaction into non-newsy months.

7.3 Why the result matters

The paper links market-wide return predictability to the micro timing of earnings announcements and challenges common assumptions about persistent expected returns and covariance-stationary monthly returns.

7.4 How to read the evidence as a whole

Table 1 establishes the earnings calendar; Tables 2–3 establish the return pattern; Table 4 and Figure 2 show real-time forecasting power; Figures 4–5 give the mechanism; Tables 7–8 test beliefs; Tables 9–10 test industry-level implications.

7.5 Limitations and cautions

The results are not a randomized experiment and may change after publication. The model abstracts from many institutional details. Still, the convergence of return, belief, industry, and international evidence makes the mechanism persuasive.

7.6 Takeaway

The main takeaway is that investors can be correct about broad earnings extrapolation but wrong about month-specific relevance. This small timing mistake can generate large, predictable return patterns.